LAST NAME: _____

Practice Quiz 08C — MATH 2055 — Fall 2026

Practice for Quiz 08

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Let $f : A \to B$ and $g : B \to C$ be functions.

   (a) Prove that if $g \circ f$ is onto, then g is onto.

   (b) Give finite sets A, B, C and functions $f : A \to B$, $g : B \to C$ for which g is onto but $g \circ f$ is not onto. Verify both claims.